

MongoDB Practice Assignment

```
test> db.users.insertOne({
|   name: "Sue",
|   age: 26,
|   status: "pending"
| });
{
  acknowledged: true,
  insertedId: ObjectId('687e0b5e843abda16c5639a8')
}
```

```
test> db.inventory.insertOne({
|   item: "canvas",
|   qty: 100,
|   tags: ["cotton"],
|   size: {
|     h: 28,
|     w: 35.5,
|     uom: "cm"
|   }
| });
{
  acknowledged: true,
  insertedId: ObjectId('687e0b67843abda16c5639a9')
}
```

```
test> db.inventory.insertMany([
|   {
|     item: "journal",
|     qty: 25,
|     tags: ["blank", "red"],
|     size: { h: 14, w: 21, uom: "cm" }
|   },
|   {
|     item: "mat",
|     qty: 85,
|     tags: ["gray"],
|     size: { h: 27.9, w: 35.5, uom: "cm" }
|   },
|   {
|     item: "mousepad",
|     qty: 25,
|     tags: ["gel", "blue"],
|     size: { h: 19, w: 22.85, uom: "cm" }
|   }
| ]);
```

```

{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('687e0b72843abda16c5639aa'),
    '1': ObjectId('687e0b72843abda16c5639ab'),
    '2': ObjectId('687e0b72843abda16c5639ac')
  }
}

```

```

test> db.users.find().pretty()
[
  {
    _id: ObjectId('687e0b5e843abda16c5639a8'),
    name: 'Sue',
    age: 26,
    status: 'pending'
  }
]

```

```

test> db.inventory.find().pretty()
[
  {
    _id: ObjectId('687e0b67843abda16c5639a9'),
    item: 'canvas',
    qty: 100,
    tags: [ 'cotton' ],
    size: { h: 28, w: 35.5, uom: 'cm' }
  },
  {
    _id: ObjectId('687e0b72843abda16c5639aa'),
    item: 'journal',
    qty: 25,
    tags: [ 'blank', 'red' ],
    size: { h: 14, w: 21, uom: 'cm' }
  },
  {
    _id: ObjectId('687e0b72843abda16c5639ab'),
    item: 'mat',
    qty: 85,
    tags: [ 'gray' ],
    size: { h: 27.9, w: 35.5, uom: 'cm' }
  },
  {
    _id: ObjectId('687e0b72843abda16c5639ac'),
    item: 'mousepad',
    qty: 25,
    tags: [ 'gel', 'blue' ],
    size: { h: 19, w: 22.85, uom: 'cm' }
  }
]

```

```

test> db.inventory.find({})
[
  {
    _id: ObjectId('687e0b67843abda16c5639a9'),
    item: 'canvas',
    qty: 100,
    tags: [ 'cotton' ],
    size: { h: 28, w: 35.5, uom: 'cm' }
  },
  {
    _id: ObjectId('687e0b72843abda16c5639aa'),
    item: 'journal',
    qty: 25,
    tags: [ 'blank', 'red' ],
    size: { h: 14, w: 21, uom: 'cm' }
  },
  {
    _id: ObjectId('687e0b72843abda16c5639ab'),
    item: 'mat',
    qty: 85,
    tags: [ 'gray' ],
    size: { h: 27.9, w: 35.5, uom: 'cm' }
  },
  {
    _id: ObjectId('687e0b72843abda16c5639ac'),
    item: 'mousepad',
    qty: 25,
    tags: [ 'gel', 'blue' ],
    size: { h: 19, w: 22.85, uom: 'cm' }
  }
]

test> db.users.find({}, { name: 1, status: 1, _id: 0 });
[ { name: 'Sue', status: 'pending' } ]

test> db.users.find({ status: "pending" })
[
  {
    _id: ObjectId('687e0b5e843abda16c5639a8'),
    name: 'Sue',
    age: 26,
    status: 'pending'
  }
]

test> db.inventory.find({}, { item: 1, qty: 1, _id: 0 })
[
  { item: 'canvas', qty: 100 },
  { item: 'journal', qty: 25 },
  { item: 'mat', qty: 85 },

```

```

    { item: 'mousepad', qty: 25 }
  ]

test> db.inventory.find({ qty: { $gt: 50 } })
[
  {
    _id: ObjectId('687e0b67843abda16c5639a9'),
    item: 'canvas',
    qty: 100,
    tags: [ 'cotton' ],
    size: { h: 28, w: 35.5, uom: 'cm' }
  },
  {
    _id: ObjectId('687e0b72843abda16c5639ab'),
    item: 'mat',
    qty: 85,
    tags: [ 'gray' ],
    size: { h: 27.9, w: 35.5, uom: 'cm' }
  }
]

test> db.inventory.find({ "size.w": { $gt: 30 } })
[
  {
    _id: ObjectId('687e0b67843abda16c5639a9'),
    item: 'canvas',
    qty: 100,
    tags: [ 'cotton' ],
    size: { h: 28, w: 35.5, uom: 'cm' }
  },
  {
    _id: ObjectId('687e0b72843abda16c5639ab'),
    item: 'mat',
    qty: 85,
    tags: [ 'gray' ],
    size: { h: 27.9, w: 35.5, uom: 'cm' }
  }
]

test> db.inventory.find({ item: { $regex: /pad/i } });
[
  {
    _id: ObjectId('687e0b72843abda16c5639ac'),
    item: 'mousepad',
    qty: 25,
    tags: [ 'gel', 'blue' ],
    size: { h: 19, w: 22.85, uom: 'cm' }
  }
]

test> db.inventory.countDocuments()
4

```

```
test> db.inventory.countDocuments({ tags: { $exists: true } })
4
```

```
test> db.users.countDocuments({ name: { $exists: true } })
1
```

```
test> db.users.countDocuments({ age: { $gt: 18 } })
1
```

```
test> db.inventory.countDocuments({ qty: { $gt: 50 } })
2
```

```
test> db.inventory.findOne()
{
  _id: ObjectId('687e0b67843abda16c5639a9'),
  item: 'canvas',
  qty: 100,
  tags: [ 'cotton' ],
  size: { h: 28, w: 35.5, uom: 'cm' }
}
```

```
test> db.users.find({ age: { $type: "int" } })
[
  {
    _id: ObjectId('687e0b5e843abda16c5639a8'),
    name: 'Sue',
    age: 26,
    status: 'pending'
  }
]
```

```
test> db.inventory.aggregate([
|   { $group: { _id: null, totalQty: { $sum: "$qty" } } }
| ]);
[ { _id: null, totalQty: 235 } ]
```

```
test> db.inventory.aggregate([
|   {
|     $project: {
|       item: 1,
|       qty: 1,
|       isHighStock: { $gt: ["$qty", 50] }
|     }
|   }
| ]);
```

```
[
  {
    _id: ObjectId('687e0b67843abda16c5639a9'),
    item: 'canvas',
    qty: 100,
    isHighStock: true
  },
  {
    _id: ObjectId('687e0b72843abda16c5639aa'),
    item: 'journal',
    qty: 25,
    isHighStock: false
  },
  {
    _id: ObjectId('687e0b72843abda16c5639ab'),
    item: 'mat',
    qty: 85,
    isHighStock: true
  },
  {
    _id: ObjectId('687e0b72843abda16c5639ac'),
    item: 'mousepad',
    qty: 25,
    isHighStock: false
  }
]
```

MongoDB installing commands for mac

Step 1: Install Homebrew

Step 2: Tap MongoDB formula

brew tap mongodb/brew

Step 3: Install MongoDB Community Edition

brew install mongodb-community

Step 4: Start MongoDB as a service

brew services start mongodb/brew/mongodb-community

Step 5: Install MongoDB Shell

brew install mongosh

Step 6: Verify Installation and version

mongosh --version